§1 经典系综

1.1 基本框架

相空间：$(q,p) = (q_1 \cdots q_{Ns}, p_1 \cdots p_{Ns})$

演化：$\begin{cases} q_i = q_i(t) \\ p_i = p_i(t) \end{cases}$, $\begin{matrix} \dot q_i = \frac{\partial H}{\partial p_i} \\ \dot p_i = -\frac{\partial H}{\partial q_i} \end{matrix}$, $H = H(q,p,t)$

态函数：$O(q,p)$

$$\frac{dO}{dt} = \sum_i \frac{\partial O}{\partial q_i}\dot q_i + \frac{\partial O}{\partial p_i}\dot p_i = \sum_i \frac{\partial O}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial O}{\partial p_i}\frac{\partial H}{\partial q_i}$$

$$= \{O, H\}$$

守恒 $\Leftrightarrow \{O, H\}$

系综：经典力学的系统的集合，每一个成员 H, N (自由度) 相同，初值相同
各成员之间独立.

$D = D(q,p,t)$，相空间中代表点的数密度 总数守恒：$\frac{dD}{dt} = 0$

$$\frac{\partial D}{\partial t} = -\frac{\partial j}{\partial q} - \frac{\partial j}{\partial p} = -\sum_i \frac{\partial}{\partial q_i}(D\dot q_i) + \frac{\partial}{\partial p_i}(D\dot p_i) \quad , i = 1\cdots Ns$$

$$= -\sum_i \frac{\partial D}{\partial q_i}\dot q_i + D\frac{\partial \dot q_i}{\partial q_i} + \frac{\partial D}{\partial p_i}\dot p_i + D\frac{\partial \dot p_i}{\partial p_i}$$

$$= -\{D, H\} + \sum_i D\frac{\partial}{\partial q_i}\left(\frac{\partial H}{\partial p_i}\right) + D\frac{\partial}{\partial p_i}\left(-\frac{\partial H}{\partial q_i}\right)$$

$$= -\{D, H\}$$

总数 $N = \int dq\, dp\, D(q,p,t) = N$

密度 $\rho := \frac{D(q,p,t)}{N}$, $\int dq\, dp\, \rho(q,p,t) = 1$

$\Leftrightarrow \frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \{\rho, H\} = 0$ Liouville Theorem

Gibbs 诠释：$\rho \rightarrow$ 概率密度.

Schordinger 方程 : $-i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{r}, t) \psi$

$\rho := \psi^* \psi$

$\vec{J} := -\frac{i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*)$

$\rightarrow \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$

密度函数 ~ 分布概率密度.

观测值 = 系综 / 概率平均.

$\langle O \rangle = \int dq dp \, O(q,p) \, \rho(q,p,t)$

$U = \langle H \rangle = \int dq dp \, H(q,p) \, \rho(q,p,t)$

$\frac{\partial \rho}{\partial t} \begin{cases} = 0 & \text{平衡统计、宏观平衡态} \quad \{\rho, H\} = 0 \\ \neq 0 & \text{非平衡.} \end{cases}$

$\{\rho, H\} = \sum_{i=1}^{N_s} \frac{\partial \rho}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial \rho}{\partial p_i} \frac{\partial H}{\partial q_i}$

$2N_s$ 值 → 需要 $2N_s$ 个独立运动积分 难.

等概率模型 $\rho(q,p) = C$

↳ 微正则系综 , 对应最大熵的.

对每个能层 : $H(q,p) = E$ 相同 → 在相空间等能面上出现概率相同

$\rho = \rho(H) = \begin{cases} C = \left(\int_E^{E+\Delta E} dq dp \right)^{-1} & H \in [E, E+\Delta E] \\ 0 \end{cases}$

$\langle O \rangle = \lim_{\Delta E \to 0} \int dq dp \, O(q,p) \, \rho(q,p,t)$